

Mingtai Zhang

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Education

Guangxi University | School of Resources, Environment and Materials

2022 – 2026

B.Eng. in Materials Science and Engineering

Coursework: physical chemistry, fundamentals of materials science, physical metallurgy, principles of metallurgical processes, materials characterization methods, electrical and electronic engineering, mechanical design.

My quantitative and computational training is self-taught. The computational work below was done with a research group at Northeastern University (Qinhuangdao), in parallel with a full experimental thesis in electrochemistry.

Research Summary

Across five projects in four fields I have held the same role: I build the part that decides whether a claim is true. In my first-author work that meant defining an attribution boundary in conversational memory, and showing that training which improves a model's replies can corrupt what it writes to a user's persistent profile, a failure reply-only evaluation never sees. Elsewhere it has meant the evaluation harness for an identifiability result in coherent imaging, and the full experimental verification of a minimax bound on estimating a selected system's accuracy from a fixed label budget. The habit began in electrochemistry, where every claim about the interface mechanism had to survive multiple independent measurement channels. I want to work on measurement design for systems that cannot be observed directly: what an evaluation has to cover, and what it costs, before a reported improvement can be believed.

Research Interests

Evaluation and auditing of LLM systems | memory, attribution, and trustworthy human–AI interaction | LLM agents for evidence-grounded reasoning | inverse problems and sample complexity | AI for experimental science

Publications and Manuscripts

Co-author lists for papers under double-blind review are withheld until notification, following the anonymity policy of each venue.

Under Review

- [1] Paper on attribution boundaries in conversational AI memory (title withheld during review) **First author**; under double-blind review, a top-tier HCI conference.
- [2] Paper on the accuracy–evaluation trade-off in test-time selection (title withheld during review) **Second author**; under double-blind review, a top-tier machine-learning conference.
- [3] Yuchen Miao, **Mingtai Zhang**, Zijun Wang, Ruobing Han, Chang Han, and Siyang Xu. “Two-Scan Recovery of Sparse Defects in Ptychography.” **Second author**; under review, *IEEE Signal Processing Letters*, 2026. (CRediT: data curation, software.)

Accepted

- [4] Yuchen Miao, Zijun Wang, Chang Han, Yurui Shi, **Mingtai Zhang**, and Siyang Xu. “The ‘10th Juror’: Open-Set Standpoint Screening for Bureaucratic Bias Detection.” *EMNLP 2026* (Main Conference) [CORE A*; CCF-B].
- [5] Guang Zhang, Dong Yang, Kewang Yang, Farwa Mushtaq, Meng Zhang, **Mingtai Zhang**, Jieshang Lu, Yucheng Wen, and Meinan Liu. “Modified Ether Electrolyte by Siloxane Additive with Abundant Si–O Bonds for High-Voltage Lithium Metal Batteries.” *Journal of Physics: Conference Series* **3165**, 012009 (2025); CEAMS 2025 [EI-indexed]. DOI: 10.1088/1742-6596/3165/1/012009.

Research Experience

Attribution and Memory in Conversational AI Agents

2026

First author, under double-blind review | problem formulation, measures, experiments, writing

- **Problem.** When a user role-plays a fictional character, a companion agent can store the character's statements as facts about the real person. I formalized *boundary non-interference*: a proposition supported only inside a fictional frame must not be asserted as a fact about the real user without attribution.
- **Two output surfaces, two measures.** Instead of scoring overall response quality, I separated the conversational channel from the persistent-memory channel, with a cross-frame leakage rate for replies and a profile-write corruption rate for generated profiles. Each came with explicit scoring rules and coverage diagnostics, plus a secondary relational-positioning construct kept clearly subordinate.
- **Dual ground truth.** Built a controlled corpus of rule-generated Chinese episodes that carry paired fictional and real-user values for the same attribute, so every error is attributable rather than inferred. Scored tens of thousands of replies and a large factorial of generated profiles across boundary cues, model scale, thinking settings, and writer prompts.
- **Main finding.** Two supervised fine-tuning variants improved conversational attribution without consistently improving profile writing, and conversation-targeted fine-tuning raised observed write corruption in every reported condition summary. Reply-level evaluation alone would have scored both interventions as successes.

Accuracy and Evaluation Cost in Test-Time Selection

2026

Second author, under double-blind review | designed and ran the entire experimental verification

- Selecting a better response also makes its accuracy harder to measure, because the favored answers are rare in the labels you already have. The paper characterizes the best accuracy reachable with N candidates when evaluation reuses L labels from the base generator, and shows that every continuous-score best-of- N rule has worst-case audit risk $\Theta(1 \wedge N/L)$, so holding precision fixed costs labels in proportion to the candidate budget.
- The two constraints, which output laws a candidate pool can reach and how hard each law is to audit, define an accuracy–audit frontier whose dual discounts correctness in regions with poor label coverage. Cross-fitted frontier policies on four frozen public reasoning banks beat uniform best-of- N mixtures at matched audit budgets, and match fully optimized order-statistic and quadratic baselines.

Sample Complexity of Sparse Recovery in Coherent Imaging

2026

Second author, IEEE SPL (under review) | forward-model simulation and the full evaluation harness

- The paper asks a minimum-measurement question: given a known uniform background carrying N unknown defects, how many scan positions are needed for unique recovery? The answer is two, and one is provably insufficient because of a conjugate-reflection ambiguity. Working on an identifiability result changed how I approach evaluation in general, in that I now ask whether the measurement can determine the answer before asking which method wins.
- I handled data curation and software: the forward-model simulation and the whole evaluation harness, including Gaussian probes, scan geometries, sparse-defect ensembles, and detector-noise injection, then benchmarked the proposed $2N$ -lag decoder against full-data linearization and the iterative phase-retrieval baselines PIE and HT-WF over defect-separation and SNR sweeps.

Open-Set Standpoint Screening with LLM Agent Panels

2026

Fifth author, EMNLP 2026 Main Conference | evaluation-set construction and experimental analysis

- Contributed evaluation-set construction and experimental analysis to MARS-Gov, which drops the closed-world assumption behind fixed bias taxonomies. A Scout proposes affected groups that no taxonomy defines, fixed and dynamic Jurors deliberate over retrieved legal and policy evidence, a Judge verifies, and a restoration agent rewrites while preserving meaning. The system reached $F1 = 0.880$, 20.2 points above the strongest zero-shot LLM, and leave-one-category-out evaluation showed that genuinely unseen categories are recoverable rather than memorized.

Controlled Electrochemistry | undergraduate thesis, Guangxi University

2025 – 2026

Sole author, advisor Prof. Meinan Liu | experimental design, three measurement channels, all analysis

- Ran a single-variable controlled study on lithium-metal anode interfaces, with every quantity fixed except one process parameter and an unmodified baseline carried through as a parallel control. With no direct image of the interface, every claim had to come from agreement between independent measurements.
- Used three cell configurations as three complementary measurement channels: half cell, symmetric cell, and full cell. A claim that survived in only one channel could not be reported as a general improvement. Plating plateau overpotential fell from 145 to 114 mV and the nucleation dip from about 30 to about 20 mV, symmetric cells stayed near 30 mV for 330 h, and full cells kept 85.3% after 600 cycles.
- Localized failure on both sides of the optimum instead of reporting only the best setting: too little treatment leaves uncovered sites and collapses at cycle 45, too much builds a layer that cracks and collapses at cycle 30. Measured the mechanism separately by impedance spectroscopy and X-ray diffraction, so the performance claim rests on an identified cause and not a correlation. Kept the whole route on commercial reagents and standard hardware so that other labs can reproduce it.

Systems Built and Deployed

Kangjie Returning-Student Tutoring Platform, Yuncheng, Shanxi | Founder

Winter 2026

- Founded and ran a non-profit tutoring service staffed by university students home for the holiday, owning scheduling, tutor–student matching, progress tracking, and daily operations.
- Rebuilt the operating workflow around AI agent tooling and automated the coordination work that had been manual, which cut the staffing cost of running the service substantially. It was also a practical test of whether agents hold up as operators rather than demos; in total the service mobilized 18 tutors to serve 37 secondary-school students.

On-Demand Fitness Coaching Platform, Guangxi University | Co-Founder

2026 – present

- Co-founded a live two-sided service with the Guangxi University Rehabilitation Association, matching certified student coaches to users for one-on-one on-site sessions. I own operating strategy, supply coordination, and the commercial model.

Industry Grounding

Sixteen manufacturing enterprises across six industries, Guangxi | four supervised placements

2024 – 2026

- Traced the aluminium chain end to end, from bauxite mining and Bayer refining through electrolysis, casting, extrusion, rolling, and lithium-battery foil, plus integrated steelmaking, automotive stamping-to-assembly, construction machinery, photovoltaic module lines, and superhard-material sintering. I have stood at the inspection points where industrial process data is actually produced, and it shapes how I think about deployed models: on a plant floor a system is judged by throughput, yield, and inspectability, not by one headline number.
- **Closest to my own work:** at **Nanning Industrial Investment Aluminium-Based New Materials Group** I followed the full ingot-to-foil chain for 12 μm lithium-battery cathode current-collector foil. The component I hand-assemble into coin cells is made there by the hundred-thousand tonne.

Leadership, Honors, and Technical Skills

Leadership: Captain, Guangxi University Football Team | Youth League Branch Secretary, University Football Association | Chair, Student Representative Assembly, School of Resources, Environment and Materials.

Honors: Silver Award, Guangxi University Career Planning Competition (2023) | Third Prize, Guangxi University Metallography Contest (2023) | First Place (2025) and Third Place (2024), Guangxi “Qianli Cup” Intercollegiate Football League, undergraduate division.

Programming & Tools: Python, PyTorch, $\text{\LaTeX}$, MATLAB, C, Claude Code, OpenClaw, Origin | **ML:** supervised fine-tuning of open-weight LLMs, large-scale batched inference, LLM-as-judge pipelines, RAG, multi-agent systems, checkpoint selection and validity filtering | **Evaluation:** factorial design, parallel controls, replication arms, holdout construction, coverage diagnostics, failure localization, pre-stated validity limits | **Signal & Imaging:** forward-model simulation, phase retrieval (PIE, HT-WF), sparse recovery, sample-complexity benchmarking | **Instrumentation:** electrochemical workstations, EIS, XRD, SEM, XPS, glovebox